Verdict (read this first)

You asked: "Build 3-4 specialised engines (BULL/BEAR/SIDEWAYS), one per regime, and a router that switches between them mid-day."

5 parallel research agents (academic literature, industry precedent, regime detection, engine training, switching mechanics) all came back with the **same answer**: that specific architecture is the most-rejected pattern in real-world quant finance. Every famous quant blow-up of the last 30 years (LTCM 1998, Quant Quake 2007, Volmageddon 2018) was a case of a single-regime model deployed into a regime it hadn't trained on.

But the **underlying instinct is right**. v5 IS leaving alpha on the table by being regime-blind in the wrong direction. The fix is just smaller and safer than 4 engines.

The right answer is two changes:

1. **Fix the detector first** (Phase 1, ~1 week). The 04-29 failure (175 blocked LONGs on a green day labelled BEAR) was caused by v5's regime detector — a static vote-aggregator with no dwell time, no hysteresis, no confidence threshold. Replace with a 2-state Gaussian HMM + hysteresis + 3-bar dwell time. This alone would have prevented yesterday's failure.

2. **Use regime as a feature, not a switch** (Phase 2, ~2 weeks). Keep the v4 LightGBM scorer. Add the regime detector's output as 3 continuous features (`P_bull`, `P_bear`, `P_sideways`). Specialise only the *exit rules and position sizing* per regime — not the scorer itself. This is exactly what Two Sigma, AQR, and the MDPI 2026 paper do.

Hard switching between specialised engines stays on the shelf as a Phase 4 option only if Phase 1+2 plateau.
